Physical Chemistry Cume

March 28, 2008

Using the *linear combinations of atomic orbitals* method with Hückel approximation, a student found out the frontier molecular orbitals (MOs), HOMO, LUMO, nHOMO, and nLUMO, or a given molecule include four MOs:

$\Phi_1 = \varphi_A + \varphi_B + \varphi_C + \varphi_D$; $\Phi_3 = \varphi_A - \varphi_B - \varphi_C + \varphi_D$

$\Phi_2 = \varphi_A - \varphi_B + \varphi_C - \varphi_D$ $\Phi_4 = \varphi_A + \varphi_B - \varphi_C - \varphi_D$

Here, φ_A, φ_B, φ_C, and φ_D are normalized AOs of four atoms in the molecule.

(1) (15 points) Please identify HOMO, LUMO, nHOMO, and nLUMO for the student. At ground state, this four-atom molecule has four valence electrons in total.

HOMO: LUMO: nHOMO: nLUMO:

(2) (20 points) If there are six electrons to be filled into these frontier orbitals, please show the correct electron configuration, assuming that the molecule is not excited. After that, please calculate the bond order for this case.

(3) (40 points) Is any of the frontier orbitals shown above normalized? If yes, please verify it (if more than one, just verify one). If not, please normalize <u>one of the MOs</u>.

(4) (25 points) For the normalized orbital in Question (3), either the one verified by you or the one normalized by you, please calculate the contribution of φ_B for this specific MO.